Unit-3
Concept Description and Association Rule Mining

Topics to be covered

- Introduction Concept description
- Data Generalization and summarization based characterization
- Attribute relevance class comparisons
- Association Rule Mining: Market basket analysis
- basic concepts,
- Finding frequent item sets: Apriori algorithm generating rules, Improved Apriori algorithm
- Incremental ARM
- Associative Classification
- Rule Mining

WHAT IS CONCEPT DESCRIPTION?

❖ Descriptive vs. predictive data mining Descriptive mining:
➢ describes concepts or task-relevant data sets in concise, summarative, informative, discriminative forms
❖ Predictive mining: Based on data and analysis, constructs models for the database, and predicts the trend and properties of unknown data
❖ Concept description:
➢ Characterization: provides a concise and succinct summarization of the given collection of data
➢ Comparison: provides descriptions comparing two or more collections of data

WHAT IS CONCEPT DESCRIPTION?

❖A concept usually refers to a collection of data such as frequent_buyers, graduate_students etc.

❖As a data mining task, concept description is not a simple enumeration (number of things done one by one) of the data.

❖Concept description generates descriptions for characterization and comparison of the data it is also called class description.

❖ Characterization provides a concise and brief summarization While of the data.

❖ concept or class comparison (also known as discrimination) provides (inequity) comparing two or more collections of data.

❖ Example

➢ discriminations Given the ABC Company database, for example, examining individual customer transactions.

➢ Sales managers may prefer to view the data generalized to higher levels, such as summarized by customer groups according to geographic regions, frequency of purchases per group and customer income.

ATTRIBUTE RELEVANCE

- Why?
  - Which dimensions should be included?
  - How high level of generalization?
  - Automatic vs. interactive Reduce # attributes; easy to understand patterns
- What?
  - statistical method for preprocessing data
    - filter out irrelevant or weakly relevant attributes
    - retain or rank the relevant attributes
  - relevance related to dimensions and levels
  - analytical characterization, analytical comparison
- How?
- Data Collection
- Preliminary
- Relevance Analysis
  - Perform AOI to removing or generalizing attributes.

ATTRIBUTE RELEVANCE

- ❖ Relevance Analysis
  - ➢ Use relevance analysis measure e.g. information gain to identify highly relevant dimensions and levels.
  - ➢ Sort and select the most relevant dimensions and levels.
- ❖ Attribute-oriented Induction for class description
  - ➢ On selected dimension/level
- ❖ OLAP operations (e.g. drilling, slicing) on relevance rules

APRIORI ALGORITHM

- With the quick growth in e-commerce applications, there is an accumulation vast quantity of data in months not in years. Data Mining, also known as Knowledge Discovery in Databases(KDD), to find anomalies, correlations, patterns, and trends to predict outcomes.

- Apriori algorithm is a classical algorithm in data mining. It is used for mining frequent itemsets and relevant association rules. It is devised to operate on a database containing a lot of transactions, for instance, items brought by customers in a store.

- It is very important for effective Market Basket Analysis and it helps the customers in purchasing their items with more ease which increases the sales of the markets. It has also been used in the field of healthcare for the detection of adverse drug reactions. It produces association rules that indicates what all combinations of medications and patient characteristics lead to ADRs.

WHAT IS AN ITEMSET?

❖A set of items together is called an itemset. If any itemset has k-items it is called a k-itemset. An itemset consists of two or more items. An itemset that occurs frequently is called a frequent itemset. Thus frequent itemset mining is a data mining technique to identify the items that often occur together.

➢ For Example, Bread and butter, Laptop and Antivirus software, etc.

WHAT IS A FREQUENT ITEMSET?

❖ A set of items is called frequent if it satisfies a minimum threshold value for support and confidence. Support shows transactions with items purchased together in a single transaction. Confidence shows transactions where the items are purchased one after the other.

❖ For frequent itemset mining method, we consider only those transactions which meet minimum threshold support and confidence requirements. Insights from these mining algorithms offer a lot of benefits, cost-cutting and improved competitive advantage.

❖ There is a tradeoff time taken to mine data and the volume of data for frequent mining. The frequent mining algorithm is an efficient algorithm to mine the hidden patterns of itemsets within a short time and less memory consumption.

FREQUENT PATTERN MINING

❖ The frequent pattern mining algorithm is one of the most important techniques of data mining to discover relationships between different items in a dataset. These relationships are represented in the form of association rules. It helps to find the irregularities in data. ❖ FPM has many applications in the field of data analysis, software bugs, cross-marketing, sale campaign analysis, market basket analysis, etc.

❖ Frequent itemsets discovered through Apriori have many applications in data mining tasks. Tasks such as finding interesting patterns in the database, finding out sequence and Mining of association rules is the most important of them.

❖ Association rules apply to supermarket transaction data, that is, to examine the customer behavior in terms of the purchased products. Association rules describe how often the items are purchased together.

WHY FREQUENT ITEMSET MINING?

❖ Frequent itemset or pattern mining is broadly used because of its wide applications in mining association rules, correlations and graph patterns constraint that is based on frequent patterns, sequential patterns, and many other data mining tasks.

APRIORI ALGORITHM

- ❖ Apriori says:
- ❖ The probability that item I is not frequent is if:
  - ➢ P(I) < minimum support threshold, then I is not frequent.
  - ➢ P (I+A) < minimum support threshold, then I+A is not frequent, where A also belongs to itemset.
  - ➢ If an itemset set has value less than minimum support then all of its supersets will also fall below min support, and thus can be ignored. This property is called the Antimonotone property.

STEPS TO FOLLOW IN APRIORI ALGORITHM

1. Join Step: This step generates (K+1) itemset from K-itemsets by joining each item with itself.
2. Prune Step: This step scans the count of each item in the database. If the candidate item does not meet minimum support, then it is regarded as infrequent and thus it is removed. This step is performed to reduce the size of the candidate itemsets.

INCREMENTAL ARM

❖ It is noted that analysis of past transaction data can provide very valuable information on customer buying behavior, and thus improve the quality of business decisions.

❖ With the increasing use of the record-based databases whose data is being continuously added, updated, deleted etc.

❖ Examples of such applications include Web log records, stock market data, grocery sales data, transactions in e-commerce, and daily weather/traffic records etc.

❖ In many applications, we would like to mine the transaction database for a fixed amount of most recent data (say, data in the last 12 months).

❖ Mining is not a one-time operation, a naive approach to solve the incremental mining problem is to re-run the mining algorithm on the updated database.

ASSOCIATIVE CLASSIFICATION

❖ Associative classification (AC) is a branch of a wide area of scientific study known as data mining. Associative classification makes use of association rule mining for extracting efficient rules, which can precisely generalize the training data set, in the rule discovery process.

❖ An associative classifier (AC) is a kind of supervised learning model that uses association rules to assign a target value. The term associative classification was coined by Bing Liu et al., in which the authors defined a model made of rules "whose right-hand side are restricted to the classification class attribute".